

Predicting Ski Jumps using State-Space Model

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Outline

- Introduction
- Modeling Framework and Dataset
 - State-space models and Least squares method
 - Data processing
- Methodology
 - Different approaches and error function
- Main Results
 - Animation app
- Discussion and Conclusion

Motivation

- ▶ Planning of ski jump enlargement and anticipating risky conditions
- ▶ Goal: Predict ski jumps based on external factors



Figure: Flying hill in Planica

- ▶ State-space models (SSM)
- ▶ Least squares method (LS)

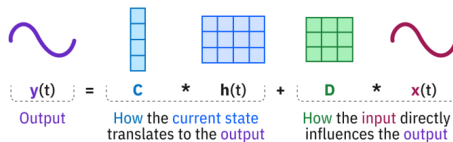
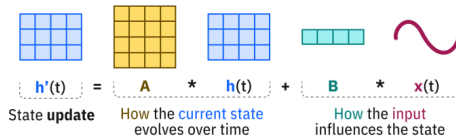


Figure: Schema of SSM

Data Description

- ▶ Cleaning the data
- ▶ Data structure
- ▶ Angles
- ▶ coordinate system on a main ski jumping hill in Planica

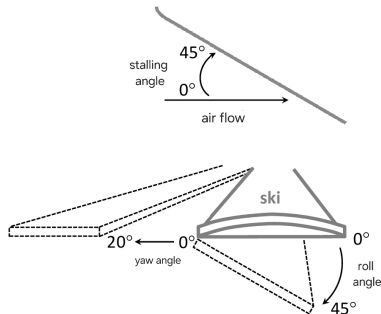


Figure: Representation of different measured angles.

Ski jumping hill

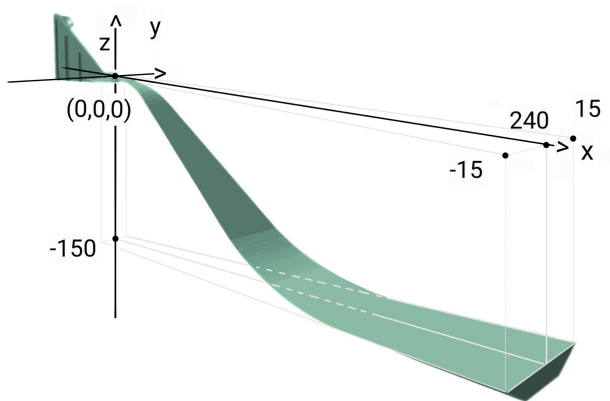


Figure: Ski jumping hill with a coordinate system placed on top.

Different approaches and error function

Different approaches:

- ▶ pure SSM
- ▶ SSM + PINN (Physics Informed Neural Networks)

Error function:

- ▶ Error function
- ▶ leave-one-out cross-validation

different add-ons to the model:

- ▶ all 12 wind sensors or only 3 wind zones
- ▶ squared wind values

Flight simulation example

Results:

- ▶ train error 1.76 m
- ▶ test error 1.84 m.

Actual vs Simulated Trajectory
Error: 1.689, Simulated length: 198.7, Actual length: 201.9

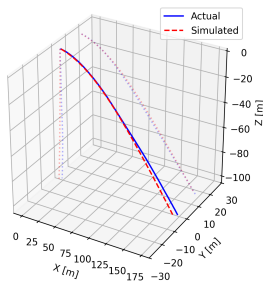


Figure: Graph with a simulated and actual flight trajectory.

Application

- Description of the application
- Observing measured and predicted jumps

Ski jump animation

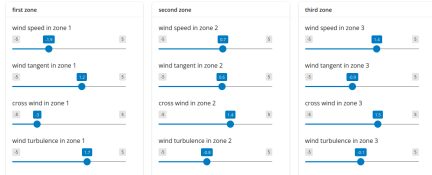
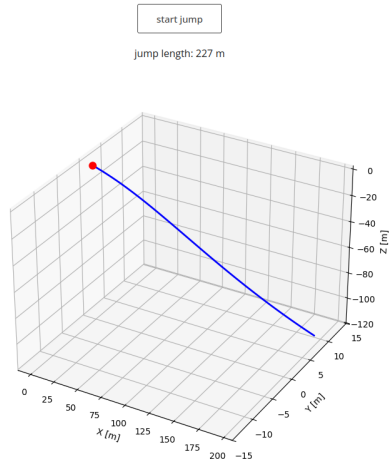


Figure: Adjusted sliders in the app



Interpretation

- ▶ Limitations
- ▶ Future work and potential improvements

Thank you!